

Global CO₂ Emissions

Trends, Regional Shifts, and Per Capita Inequalities

Scope: 1750 – 2024 | Data: Global Carbon Budget 2025, NASA

A data-driven overview of how global carbon dioxide emissions have evolved from the Industrial Revolution to 2024, highlighting regional shifts, per capita disparities, and the latest annual change trajectories.

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

Mauna Loa Observatory, Feb 2026

↑ 50% from ~280 ppm pre-industrial baseline

Context

280 ppm

Pre-industrial baseline (~1750)

310 ppm

Post-WWII recovery (1950)

353 ppm

Keeling Curve start (1958)

421 ppm

2023 — first year above 420 ppm

429 ppm

February 2026 — latest reading

Source: NOAA Mauna Loa Observatory & NASA Climate — ground-based and satellite measurements confirm sharp acceleration linked to coal, oil, and natural gas combustion.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Tonnes



Reference: figure_2.jpg — Global CO₂ emissions trajectory (Global Carbon Budget 2025)

Key Facts

- Fossil fuel & industry emissions
- Growth rate slowed but emissions have NOT yet peaked
- Coal remains the largest single source
- 2024 total: 35–37 billion tonnes CO₂

China Now Emits Over 12 Billion Tonnes — More Than Double the United States

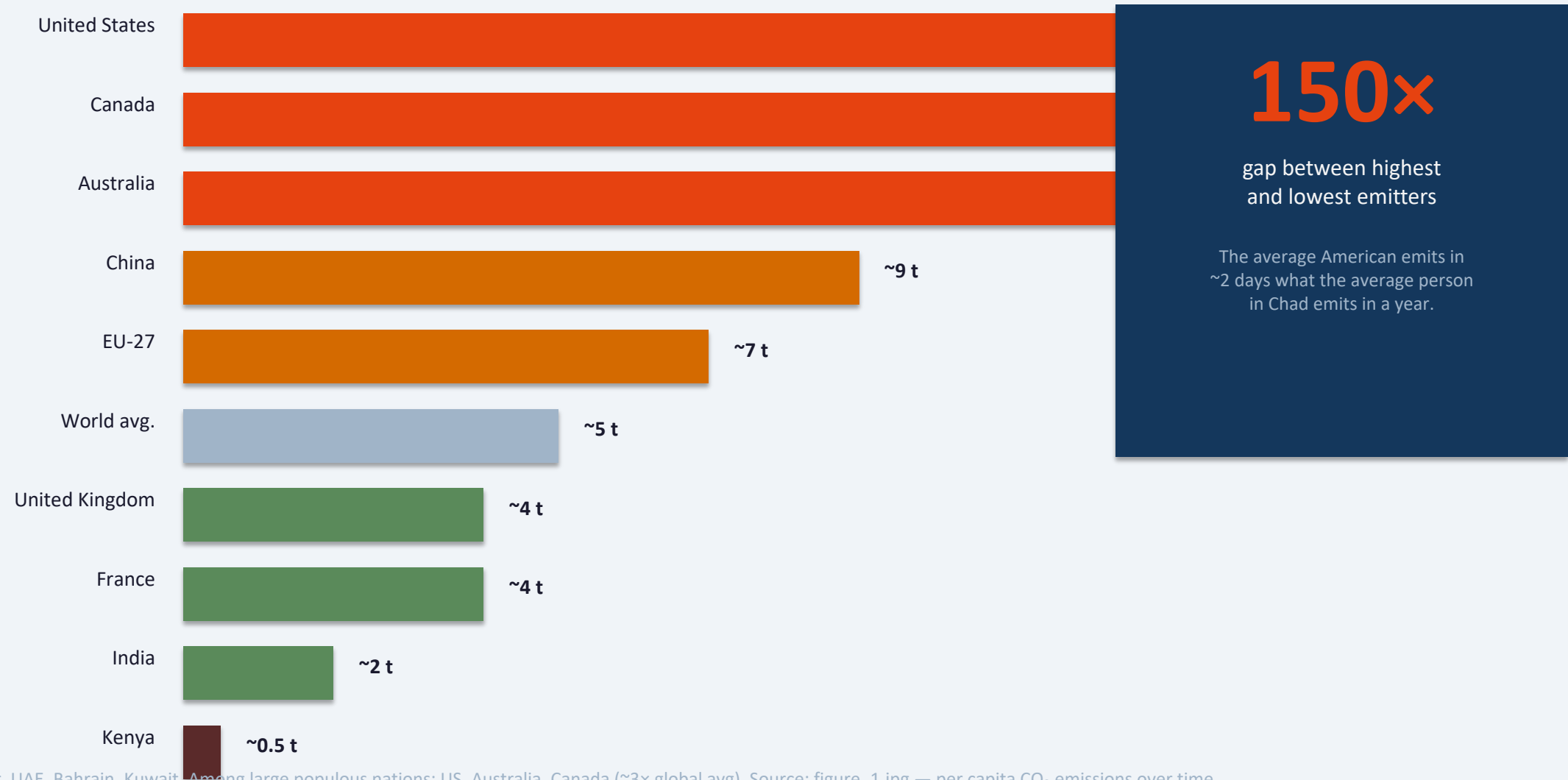
Country / Region	Approx. Annual CO ₂ (Bt)	Share of Global	Trend
China	~12.5	~27%	Slight increase ↑
United States	~5.0	~14%	Declining ↓
India	~3.0	~8%	Rising fast ↑↑
EU-27	~2.7	~7%	Declining ↓
Germany	~0.6	~1.7%	Declining ↓
United Kingdom	~0.3	~0.9%	Declining ↓
Brazil	~0.5	~1.4%	Stable →

 US peaked near 6 Bt (~2005); now ~5 Bt. China post-2000 rise steep, now >25% of global total. India growing fastest among major emitters.

Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One-Third



Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



Highest per capita: Qatar, UAE, Bahrain, Kuwait. Among large populous nations: US, Australia, Canada (~3× global avg). Source: figure_1.jpg — per capita CO2 emissions over time.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase



Declining: -1% to -5%



Moderate rise: +2% to +5%



Sharp rise: +5% to +10%+

▼ Declining / Blue on Map

- **United States** [↓ ~-2% to -4%]

US emissions declining but still ~5 Bt/yr

- **Much of Europe** [↓ ~-1% to -3%]

Germany, UK, France all down; coal phase-outs

- **Parts of Latin America** [↓ small]

Brazil stable; one or two countries saw larger drops

▲ Rising / Orange-Red on Map

- **Russia & former USSR** [↑ +2% to +7%]

Higher fossil fuel use; war-related energy shifts

- **Southeast Asia** [↑ +3% to +8%]

Vietnam, Philippines, Indonesia driving growth

- **Parts of Africa** [↑ +2% to +10%]

Limited clean energy access; growing coal use

- **Some S. American** [↑ >10%]

One-off baseline effects; close monitoring needed

Reference: figure_4.jpg — Annual percentage change in CO₂ emissions map, 2024. Mixed global picture: structural declines in wealthy nations vs. rapid growth in emerging economies.

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar living standards can have very different per capita CO₂ emissions depending on their energy source mix. Policy and infrastructure choices matter as much as prosperity.

	France	United Kingdom	Germany	Netherlands
Per Capita CO ₂ (t/yr)	~4	~4	~8	~7.5
Nuclear / Renewables share	~80%	~45–50%	~50%	~30%
Fossil fuel dependency	~15–20%	~50–55%	~50–55%	~65–70%
Key energy sources	Nuclear, hydro, wind, solar	Wind, gas, nuclear, North Sea oil	Wind, solar, gas, coal (declining)	Gas, wind, solar, some coal

📌 France's heavy nuclear investment (追溯到 1970s oil shocks) keeps per capita emissions at ~4 t — half Germany's ~8 t — despite similar GDP per capita. The UK has cut coal sharply since 2012, driving recent declines. This demonstrates that energy policy is a climate lever, not just a consequence of wealth.

Key Takeaways

Emissions Are Still Rising — But the Map of Responsibility Is Shifting

1 Global emissions exceed 35 Bt/year and have NOT yet peaked.

Despite slowing growth rates, fossil fuel CO₂ continues to accumulate. The peak is political, not yet physical.

2 China alone accounts for >25% of global total (~12.5 Bt).

More than double the US (~5 Bt); India (~3 Bt) is the third-largest and rising fastest. The geographic center is now firmly in Asia.

3 Per capita inequality remains extreme — a 150× gap.

US/Australia (~15 t/person) vs. Chad/Niger (~0.1 t/person). The average American emits in two days what a Chadian emits in a year.

4 US and Europe are declining in annual terms, but historical cumulative responsibility remains large.

Both regions have contributed the majority of all historical CO₂. Per capita emissions in wealthy nations still far exceed global averages.

5 Energy mix and policy choices — not just prosperity — determine per capita outcomes.

France (~4 t) vs. Germany (~8 t) demonstrates that nuclear and renewable-heavy energy systems can achieve low per capita emissions at similar living standards.